Task 1:

Creating an Outbound Firewall Rule to Block Spotify App

Objective: Configure Windows Defender Firewall with Advanced Security to create a custom Outbound Rule that blocks all outgoing network traffic from the Spotify desktop application.

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1. Step-by-Step Configuration Guide:

1.Open Firewall Advanced Security Console: Step 1.

Press Win + R, type wf.msc in the Run dialog box, and press Enter to launch Windows Defender Firewall with Advanced Security.

2.Initiate New Outbound Rule Wizard: Step 2.

In the left navigation pane, right-click on Outbound Rules and select New Rule... to open the New Outbound Rule Wizard.

3.Select Rule Type: Step 3.

Choose the Program rule type option and click Next.

4.Specify Program Path: Step 4.

Select This program path, click Browse, and navigate to the Spotify executable path (e.g., %APPDATA%\Spotify\Spotify.exe or C:\Users\<Username>\AppData\Roaming\Spotify\Spotify.exe), then click Next.

5.Specify Action: Step 5.

Select Block the connection and click Next.

6.Select Profile Scope: Step 6.

Check all three network profile options (Domain, Private, and Public) to enforce the block across all network environments, then click Next.

7.Name and Apply Rule: Step 7.

Enter the Rule Name as Block_Spotify_Outbound and click Finish to activate the rule.

2. Verification Method:

To verify that the rule is active, open the Spotify Desktop App and attempt to play an online track. Spotify will display an offline error message ("No internet connection detected"), confirming that outgoing traffic is successfully blocked by the firewall rule.

Task 2:

Creating an Inbound Firewall Rule for Google Chrome (Port 443 HTTPS)

Objective: Configure Windows Defender Firewall with Advanced Security to create a custom Inbound Rule allowing incoming connections exclusively for the Google Chrome application on TCP Port 443 (HTTPS protocol).

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1. Step-by-Step Configuration Guide:

1.Open Firewall Advanced Security Console: Step 1.

Press Win + R, type wf.msc in the Run dialog box, and press Enter to launch Windows Defender Firewall with Advanced Security.

2.Initiate New Inbound Rule Wizard: Step 2.

In the left navigation pane, right-click on Inbound Rules and select New Rule... to launch the New Inbound Rule Wizard.

3.Select Rule Type: Step 3.

Select Program as the rule type and click Next.

4.Specify Application Path: Step 4.

Select This program path and enter or browse to Google Chrome's path:

C:\Program Files\Google\Chrome\Application\chrome.exe (or C:\Program Files (x86)\Google\Chrome\Application\chrome.exe), then click Next.

5.Configure Action: Step 5.

Select Allow the connection and click Next.

6.Configure Protocol and Port Settings: Step 6.

In the newly created rule properties or wizard customization, select Protocols and Ports:

- Protocol type: TCP
- Local port: Specific Ports -> 443

7.Apply Profile & Name Rule: Step 7.

Select all profiles (Domain, Private, and Public), click Next, and set the Name as Allow_Chrome_Inbound_HTTPS_443, then click Finish.

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2. Verification Method:

To verify the rule active state:

1. Open PowerShell / Command Prompt as Administrator and run:

netsh advfirewall firewall show rule name="Allow_Chrome_Inbound_HTTPS_443"

Confirm that the status displays **Enabled: YES**, **Action: Allow**, **Protocol: TCP**, and **LocalPort: 443**.

Task 3:

Blocking Traffic from a Specific IP Address Using Firewall Scope Tab

Objective: Document a step-by-step procedure to create a custom firewall rule in Windows Defender Firewall that blocks all network communications (inbound and outbound) to/from a target IP address (123.45.67.89).

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1. Step-by-Step Configuration Guide:

1.Open Advanced Firewall Management Console: Step 1.

Press Win + R, type wf.msc in the Run window, and press Enter to launch Windows Defender Firewall with Advanced Security.

2.Start New Rule Wizard: Step 2.

Right-click on Inbound Rules (or Outbound Rules depending on traffic direction) in the left-hand menu and click New Rule....

3.Select Custom Rule Type: Step 3.

In the Rule Type step, choose Custom and click Next.

4.Program & Protocol Selection: Step 4.

Leave All programs selected and click Next. Keep Protocol type: Any and click Next.

5.Configure IP Address Filter (Scope Tab): Step 5.

Under the Which remote IP addresses does this rule apply to? section:

1. Select These IP addresses:
2. Click the Add... button.
3. Choose This IP address or subnet: and enter 123.45.67.89.
4. Click OK and then click Next.

6.Set Block Action: Step 6.

Select Block the connection and click Next.

7.Apply Profiles & Name Rule: Step 7.

Ensure all network profiles (Domain, Private, Public) are checked, click Next, set the name as Block_IP_123.45.67.89, and click Finish.

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2. Verification Command:

To verify that the IP restriction rule has been created and activated, open Command Prompt as Administrator and run:

netsh advfirewall firewall show rule name="Block_IP_123.45.67.89"

Expected Output: Displays Enabled: YES, Action: Block, and RemoteIP: 123.45.67.89/32.

Task 4:

Comparative Analysis: Hardware Firewalls vs. Software Firewalls

Objective: Compare Hardware Firewalls (Enterprise Network Appliances) and Software Firewalls (Host-Based Security Applications) by evaluating 3 fundamental operational differences and providing specific practical use cases for each.

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1. Hardware Firewall vs. Software Firewall Comparison Table:

Comparison Parameter	**Hardware Firewall (Standalone Appliance)**	**Software Firewall (Host-Based Application)**
1. Deployment & Infrastructure Level	Physical, dedicated security appliance installed at the perimeter	Software application/service installed locally on an individual host operating system

Comparison Parameter	Hardware Firewall (Standalone Appliance)	Software Firewall (Host-Based Application)
	boundary (gateway) between the internal private network and the external internet.	(e.g., Windows Defender, iptables).
2. Performance & System Resource Impact	Runs on specialized dedicated hardware processors (ASIC/NPU). Processes high volume traffic without consuming endpoint host CPU or RAM.	Relies entirely on the host computer's CPU, Memory, and System resources, which can impact device speed during heavy traffic inspection.

Comparison Parameter	Hardware Firewall (Standalone Appliance)	Software Firewall (Host-Based Application)
3. Protection Scope & Traffic Inspection	Protects the entire network perimeter and all connected devices (servers, PCs, IoT, Wi-Fi clients) in a single centralized location.	Protects only the specific individual host device where the software firewall is installed and running.

2. Practical Use Cases:

- **Hardware Firewall Use Case:**
    - **Enterprise / Campus Network Perimeter Security:** Used by large corporate offices, enterprise data centers, and banks at the edge router to filter incoming malicious web traffic,

block unauthorized intrusions, and manage VPN connections for hundreds of connected devices simultaneously.

- **Software Firewall Use Case:**
    - **Individual Endpoint & Remote Worker Protection:** Used on personal laptops, remote worker PCs, and mobile endpoints (e.g., Windows Defender Firewall) to protect the device when connecting to untrusted public Wi-Fi networks (cafes, airports) where network-level hardware firewalls are unavailable.